Claude's sessions: RT by trial, transitivity and consistency

Data. Claude's raw session CSVs (read directly, bypassing preprocessing) and the human pilot reference from data/processed/cbcu.csv (10 participants x 2 sessions x 105 trials).

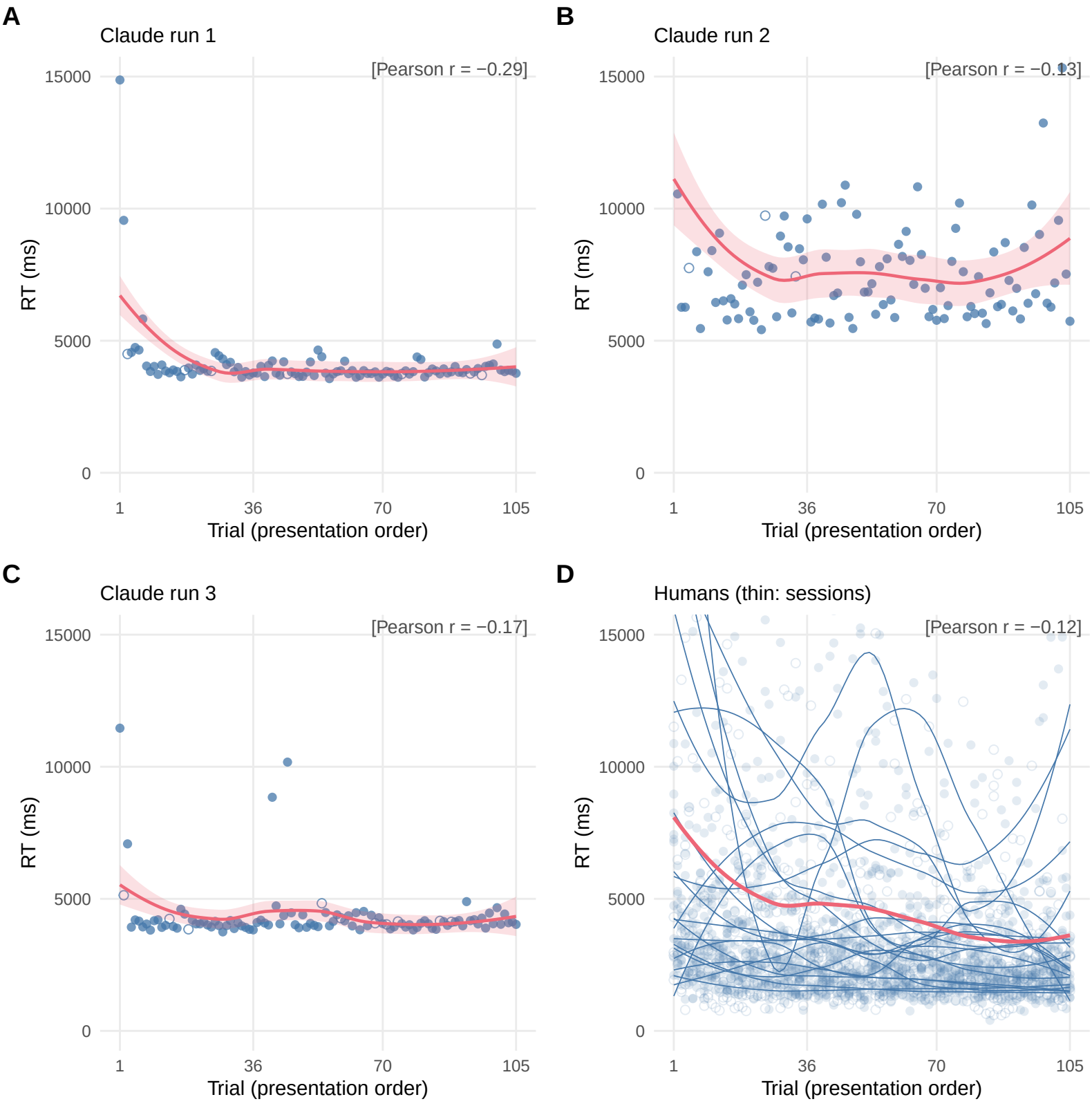
Claude run 1 (PHQ9_pairwise_kpd7ta7l_SESSION_2026−10−09_11h00.23.625.csv): 242 raw rows, 105 choice rows kept
Claude run 2 (PHQ9_pairwise_spf06mb2_SESSION_2026−10−12_10h23.05.926.csv): 242 raw rows, 105 choice rows kept
Claude run 3 (PHQ9_pairwise_uz6b41mu_SESSION_2026−10−12_12h51.19.213.csv): 242 raw rows, 105 choice rows kept

Only rows with trial_type html−button−response and phase pairwise are kept; all other rows are dropped (row types of run 1 shown; run 2 has the same structure):

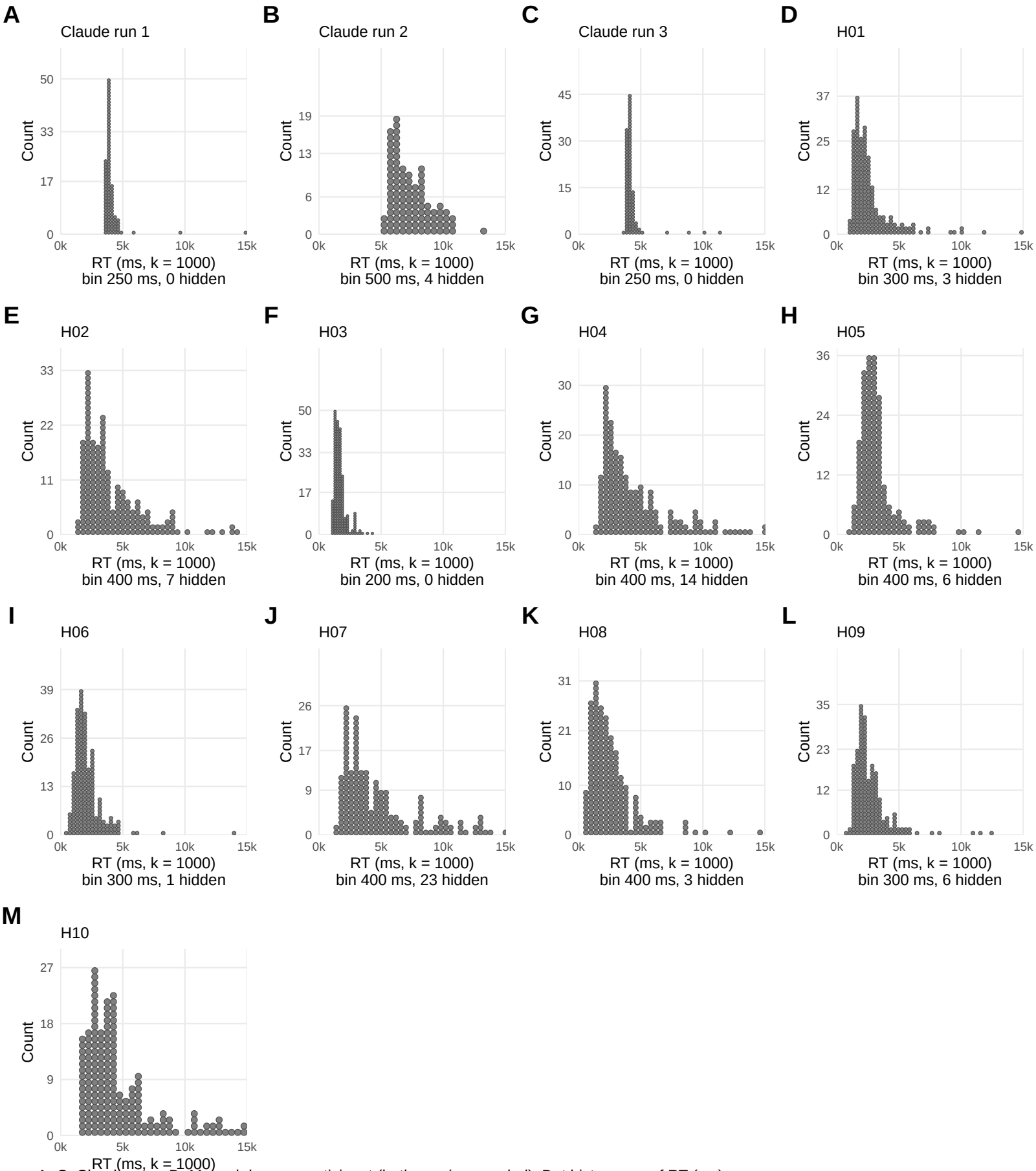
trial_type	phase	rows	kept
fullscreen	(none)	1	dropped
html−button−response	pairwise	105	KEPT
html−button−response	phase2_break	3	dropped
html−button−response	phase2_instructions	7	dropped
html−button−response	phase2_ready	1	dropped
html−button−response	phq9_grid	1	dropped
html−button−response	quiz	6	dropped
html−button−response	(none)	7	dropped
html−keyboard−response	iti	105	dropped
html−keyboard−response	(none)	3	dropped
pavlovia	(none)	1	dropped
survey−text	feedback	1	dropped
survey−text	free_text_explanation	1	dropped

Checks that passed (the script stops if any fails): exactly 105 choice rows; trial order 1−105 without duplicates; no repeated pair; no missing RT; window_status 'ok' on every row; choice values left/right/skip; skip matches the skipped flag; chosen item matches the chosen side. No RT outliers were removed.

Human reference: 20 participant−sessions, 105 trials each (checked).
RT axis cap in the figure: 15,000 ms (trials above it are hidden, not removed):
Claude: 4 of 315 trials above 15000 ms
Human: 87 of 2100 trials above 15000 ms



A–C: Claude runs, RT (ms) against presentation order, LOESS line with band, hollow points are skipped trials.
D: human sessions, one thin LOESS line per session, pooled line in rose.



A–C: Claude runs. D–M: each human participant (both sessions pooled). Dot histograms of RT (ms), same x-axis in every panel, cut at 15,000 ms; 'hidden' = trials above the cap (not removed).

Transitivity and consistency

One row per participant−session. Skip trials are ignored for transitivity and consistency.
What each column means is explained on the next page. NA = no decided choice (H08 skipped every trial). Participants with many skips have few decided triples, so their values are unstable.

unit	n_decided	skip_rate	n_decided_triples	n_cyclic	cycle_rate	min_contradictions
Claude_run1	98	0.067	375	0	0.000	0
Claude_run2	102	0.029	418	0	0.000	0
Claude_run3	95	0.095	345	0	0.000	0
H01_time1	99	0.057	385	1	0.003	1
H01_time2	99	0.057	385	0	0.000	0
H02_time1	104	0.010	442	10	0.023	4
H02_time2	104	0.010	442	10	0.023	4
H03_time1	26	0.752	12	0	0.000	0
H03_time2	27	0.743	13	0	0.000	0
H04_time1	85	0.190	260	14	0.054	4
H04_time2	92	0.124	318	13	0.041	3
H05_time1	76	0.276	198	1	0.005	1
H05_time2	70	0.333	161	6	0.037	2
H06_time1	103	0.019	430	21	0.049	6
H06_time2	105	0.000	455	62	0.136	13
H07_time1	28	0.733	17	0	0.000	0
H07_time2	36	0.657	32	1	0.031	1
H08_time1	0	1.000	0	0	NA	NA
H08_time2	0	1.000	0	0	NA	NA
H09_time1	86	0.181	269	0	0.000	0
H09_time2	81	0.229	232	16	0.069	5
H10_time1	104	0.010	442	15	0.034	6
H10_time2	105	0.000	455	18	0.040	7

Human range (min / median / max) next to Claude

measure	min	median	max	Claude_run1	Claude_run2	Claude_run3
skip_rate	0	0.186	1.000	0.067	0.029	0.095
cycle_rate	0	0.027	0.136	0.000	0.000	0.000
min_contradictions	0	2.500	13.000	0.000	0.000	0.000

What each column means

unit
One participant in one session. Claude_run1 and Claude_run2 are Claude's two sessions; H01_time1 is human participant 1 in session 1, H01_time2 is the same participant in session 2.

n_decided
Trials, out of 105, where an item was actually chosen (left or right, not skip).

skip_rate
Share of the 105 trials answered with 'neither bothered me' (skip).

n_decided_triples
A triple is a group of three items (A, B, C). With 15 items there are 455 of them: 15 x 14 x 13 = 2,730 ordered ways, divided by 6 because the order does not matter. A triple can only be checked for a contradiction if all three of its pairs (A vs B, B vs C, A vs C) were decided. This is how many triples could be checked.

n_cyclic
How many of those triples are circular: A chosen over B, B over C, but C over A. No single order of the three items explains these choices, so each one is a violation of transitivity.

cycle_rate
n_cyclic / n_decided_triples: the share of the checkable triples that are circular. 0 = perfectly consistent (transitive); higher = more contradictions. Example: H06 in session 2 skipped nothing, 62 of 455 triples were circular, so 62 / 455 = 0.136. A random chooser is around 0.25. Skipped pairs shrink the denominator, so a participant with many skips is judged on few triples (e.g. about 12 for H03) and the value is less reliable.

min_contradictions
The fewest choices that go against the best single ranking of the 15 items (found exactly). 0 means one ranking explains every choice; a higher number means noisier choices. Example: H06 in session 2 has 13. NA when the participant made no decided choice.